SOC AI Search: Trustworthy Natural-Language Queries for SOC Triage

CSC13114 | PA#1 - Project Proposal & Planning

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Repository: https://github.com/PhamVanMinhBinhThuan/HCMUS-AWAD-SOC-AI-SEARCH

1. Problem and Users

Mai, a *fictional Tier-1 SOC analyst* on an overnight shift, investigates repeated VPN login failures. She currently copies and edits Elasticsearch Query DSL, risking incorrect fields, filters or time windows. Our console accepts English/Vietnamese questions and shows exactly what it will search.

2. One Core LLM Feature and the Cost of Being Wrong

The core feature is **natural-language question -> validated SearchPlan -> Elasticsearch results**. For "Show failed logins in the last 24 hours", the LLM proposes a constrained plan; Spring Boot validates and deterministically compiles it, then shows **Query Breakdown** before execution. The LLM never submits DSL; ambiguous high-impact conditions require confirmation.

A valid plan may silently use `now-1h` instead of `now-24h`. In a planned fixture with 60 relevant events, this omits 40 (66.7% recall loss), potentially falsely reassuring Mai and delaying escalation. She can rerun it, but cannot recover the delay.

We will test 120 bilingual questions against reference plans and results. Targets: **90% critical-filter exact match**, **95% event recall**, **zero disallowed executions**, and clarification for ambiguous high-impact requests.

3. Scope for This Semester

In: Bilingual React console; validated SearchPlan-to-Elasticsearch execution; 10,000 synthetic events; fixed analyst/viewer login; Query Breakdown correction/clarification; search, count, top-N and time-series queries; reproducible evaluation.

Out: Real data, live SIEM integration, automated response, fine-tuning, autonomous investigations, dashboards, summaries, suggestions, exports and role administration.

4. Six Checkpoints: Work, Owner and Date

Checkpoint	Date	Owner and deliverable
CP1 Foundations	**2026-10-26**	Minh: repo, Docker/CI, schema; Khoa: data and 20 checked questions
CP2 Core path	**2026-11-09**	Minh: validator, compiler, Gemini/mock; Khoa: console and Query Breakdown
CP3 Data/oracle	**2026-11-23**	Khoa: 120 questions; Minh: oracle, login, query log; both: review
CP4 Evaluation	**2026-12-07**	Khoa: scoring/baseline; Minh: confirmation gate
CP5 Validation	**2026-12-21**	Khoa: correction UI/evaluation; Minh: regression/correction tests
CP6 Delivery	**2027-01-04**	Minh: deployment/demo; Khoa: cost report/final comparison

Minh = Phạm Văn Minh; Khoa = Trần Anh Khoa. Dependencies are sequential; both review the labels.

5. Two Project-Ending Risks and Immediate Mitigations

1. **Wrong but valid queries** could hide threats. This week: test 20 authored questions with Gemini, log mismatches and require confirmation for ambiguous high-impact requests.
2. **Unreliable ground truth** could invalidate evaluation. This week: freeze the generator, check 20 reference queries and run an LLM-independent oracle.

6. Technology Choices and Budget

Choice	Why this project needs it
React/TypeScript	Bilingual console and editable Query Breakdown.
Java 21/Spring Boot	Plan validation, authorization and deterministic DSL.
Elasticsearch/PostgreSQL/Keycloak	Log search, audit history and fixed-role access.
Docker Compose/GitHub Actions	Reproducible stack and evaluation.
Gemini 3.1 Flash-Lite	Affordable structured plans; deterministic CI mock.

LLM cost: Google pricing gives an estimated **USD 0.00075/request**, or **USD 1.50 per 2,000 calls**; we will log usage and cap the initial budget at **USD 10**.